Supp I

First, $(\mathbb{Z}/5\mathbb{Z})^2$ contains $25 = 5 \cdot 5$ vectors, since by definition it is just ordered pairs of elements of $\mathbb{Z}/5\mathbb{Z}$, of which there are 5.

How many unordered bases does $(\mathbb{Z}/5\mathbb{Z})^2$ have?

Two steps:

a) show that unordered bases of $(\mathbb{Z}/5\mathbb{Z})^2$ correspond to unordered pairs of linearly independent vectors. (ie show that all such span)

b) count them up.

pf (a). Linearly independent pairs are just pairs where one is not a scalar multiple of the other. If we have this in $\mathbb{F}^2$, then they also span. Just play them off of each other (ie take linear combinations) to get the standard basis, which of course spans. (More generally, we will formalize this using row reduction). Similarly, any more than two vectors will be linearly dependent in $\mathbb{F}^2$, since if the first two vectors are linearly independent, we've just shown that they together already span $\mathbb{F}^2$ and so tossing in any third vector yields a linearly dependent set.

Pf (b) The vectors in each pair that we will count will be distinct (since they are not ~~linear~~ scalar multiples of each other).
So we will count ordered pairs, then divide by 2.

How many choices for first vector?
By Supp IV, we can take any nonzero vector in $(\mathbb{Z}/5\mathbb{Z})^2$.

So there are $25-1=24$ choices.

Given a choice of a first vector v_1, how many second vectors are possible?
We know that it must not be a scalar multiple of v_1 (of which there are exactly 5), and any of the remaining vectors will work.
So there are $25-5=20$ choices for the second vector.

So # ordered bases is $24 \cdot 20 = 480$

and # unorderd bases is $\frac{24 \cdot 20}{2} = \boxed{240}$.

Finally, iterating this idea yields the calculation:

$$\frac{(p^n-1)(p^n-p)(p^n-p^2)\cdots(p^n-p^{n-1})}{n!}$$

Supp II (answers only) (idea of proof, but not proof)

1. $\boxed{N}$ (proper subset of B_1, so can't span)

2. $\boxed{N}$ (B_1 is a proper subset, so can't be lin ind)

3. $\boxed{N}$ (proper subset of B_1, so can't span)

4. $\boxed{A}$ (contains B_1, so spans V)

Supp III: pairs of vectors in V J & K s.t. $J \cap K = \emptyset$.

1. $\text{span}(J) = \text{span}(K)$ $\boxed{S}$

example where true:

$J = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$, $K = \left\{ \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\}$

example where false:

$J = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \end{pmatrix} \right\}$, $K = \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\}$

2. $\text{span}(J)$ and $\text{span}(K)$ are disjoint $\boxed{N}$

The zero vector is in any span.
Just take the trivial linear combination of the vectors.
So $0 \in \text{span}(J)$ and $0 \in \text{span}(K)$
so $\text{span}(J)$ and $\text{span}(K)$ are not disjoint.

3. If there exists a vector x where $\text{span}(J \cup \{x\}) = \text{span}(K \cup \{x\})$, then $\text{span}(J) = \text{span}(K)$. $\boxed{S}$

(i.e. in general there is no "cancellation property" for sets and their spans.)

example where true:

If $\text{span}(J) = \text{span}(K)$ to start with, then any choice of x will work.
That is, adding the same new vector to two sets with equal spans will keep their spans equal.

example where false:

for instance, maybe x is redundant for one of J or K...

$J = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \end{pmatrix} \right\}$ $K = \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right\}$

and let $x = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

Supp IV

One approach is to follow our proof that every v.s. with a finite spanning set has a finite basis.

Let $v \in V$ s.t. $v \neq 0$

Let S be a finite spanning set for V.

Then $S \cup \{v\}$ is also a spanning set.

Want to iteratively remove vectors from $S \cup \{v\}$ — without ever removing v — so that the resulting set still spans, and so that we take steps toward linear independence.

There is a little bit of case analysis to show that what we can remove during a give step can always be chosen to not be v.

Supp V. Note: If $AB=BA$, then either A & B are both $n\times n$, or (more broadly) A is $n\times m$ and B is $m\times n$.

~~What comm~~

For this problem, we'll just discuss the square case, and focus on 2×2. But more general things apply.

What commutes with $S=\begin{pmatrix}1&1\\0&1\end{pmatrix}$?

For any $n\times n$ matrix, A, two elements that commute with A are guaranteed: A itself, and the $n\times n$ identity matrix I

$$\cancel{A\cdot A = AA} \qquad AA \overset{\checkmark}{=} AA \qquad AI = A = IA \checkmark$$

So in our case we can $A=I=\begin{pmatrix}1&0\\0&1\end{pmatrix}$ and $B=S=\begin{pmatrix}1&1\\0&1\end{pmatrix}$ as two matrices that commute with S.

Do all linear combinations of A and B commute with S? Let's check, in general.

$$S(kA+lB)=\begin{pmatrix}1&1\\0&1\end{pmatrix}\left[\begin{pmatrix}k&0\\0&k\end{pmatrix}+\begin{pmatrix}l&l\\0&l\end{pmatrix}\right]=\begin{pmatrix}1&1\\0&1\end{pmatrix}\begin{pmatrix}k+l&l\\0&k+l\end{pmatrix}$$
$$=\begin{pmatrix}k+l&k+2l\\0&k+l\end{pmatrix}$$

and $$(kA+lB)S=\begin{pmatrix}k+l&l\\0&k+l\end{pmatrix}\begin{pmatrix}1&1\\0&1\end{pmatrix}=\begin{pmatrix}k+l&k+2l\\0&k+l\end{pmatrix}$$

They agree! So the answer to the first part (for this choice of A & B) is yes.

Supp V (continued)

Question: what are <u>all</u> matrices that commute with $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$? Want: $\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & a+b \\ c & c+d \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a+c & b+d \\ c & d \end{pmatrix}$$

Equating entries yields:
$$\begin{aligned} a &= a+c \\ c &= c \\ a+b &= b+d \\ c+d &= d \end{aligned}$$
(A system of linear equations!)

$\Rightarrow$ $\boxed{c=0}$ and $a+b = b+d$ or $\boxed{a=d}$

So the matrices that commute with $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ are the matrices of the form $\begin{pmatrix} a & b \\ 0 & a \end{pmatrix}$

Note: This is a v.s.! A 2-dim subspace of $M_{2\times 2}^{\mathbb{R}}$ (really $M_{2\times 2}^{\mathbb{F}}$)

Also, this shows that $\text{span}\left(\left\{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\right\}\right)$ is equal to the set of all matrices that commute with $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

Supp V (cont, 2)

Finally, applying some properties of matrix multiplication from Ch. 1.5.3, we have:

Suppose ~~AB=BA~~ $AC = CA$ and $BC = CB$.

Then

$$C(kA + \ell B) \overset{\text{distrib}}{=} C(kA) + C(\ell B)$$

$$\overset{\text{scalars}}{=} (kC)A + (\ell C)B$$

$$\overset{\text{reassoc}}{=} k(CA) + \ell(CB)$$

$$\overset{\text{apply the given}}{=} k(AC) + \ell(BC)$$

Now undo the other steps:

$$\overset{\text{scalars}}{=} (kA)C + (\ell B)C$$

$$\overset{\text{distrib}}{=} (kA + \ell B)C \qquad \blacksquare$$

1.2.6

If v_1 v_2 v_3 are linearly dependent,

can $w_1 = v_1 + v_2$, $w_2 = v_2 + v_3$, $w_3 = v_3 + v_1$

be linearly <u>independent</u>?

No. Intuitively, the v_i must live in a lower-dim subspace. But then so do the w_i, since they are linear combos of the v_i. But if the w_i live in a lower-dim subspace, then they aren't linearly independent.

As I mentioned in the announcement, we will develop more powerful tools that will later allow us to answer this kind of question in a straightforward way.

1.3.6. Identify $\mathbb{C}$ with $\mathbb{R}^2$:

$$z = x + iy$$

$$f : \mathbb{C}^1 \to \mathbb{R}^2$$

$$z \quad \begin{pmatrix} x \\ y \end{pmatrix}$$

just a function, not a linear transformation (the v.s's are over different fields)

a) multiplication in $\mathbb{C}$ is a linear transformation on the v.s. $\mathbb{C}^1$:

$$(\alpha)(z) = (\alpha z)$$

← result of multiplying matrix times vector

1×1 matrix, representing a linear transformation $\mathbb{C}^1 \to \mathbb{C}^1$

element of v.s. $\mathbb{C}^1$

b) Want: linear transformation $\mathbb{R}^2 \to \mathbb{R}^2$ such that $(\alpha)(z) = (\alpha z)$ aka

$$\begin{pmatrix} - & - \\ - & - \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} =$$

$$(a+bi)(x+iy) =$$

$$ax + iay + bxi - by$$

$$\begin{pmatrix} x \\ y \end{pmatrix} \longmapsto \begin{pmatrix} ax - by \\ ay + bx \end{pmatrix} \qquad \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

c) Note that every complex linear transformation of $\mathbb{C}^1$ preserves angles, whereas this is not true for linear transformations of $\mathbb{R}^2$; these have a greater degree of freedom.